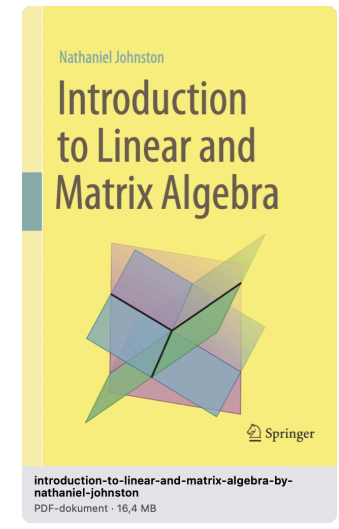
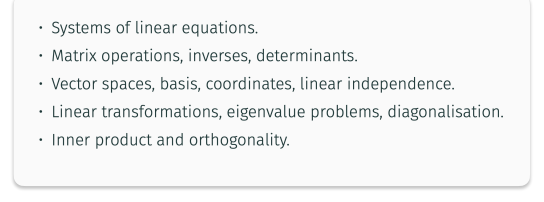
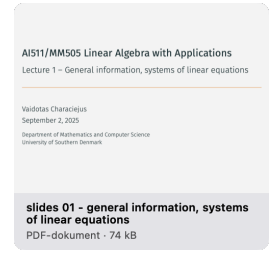
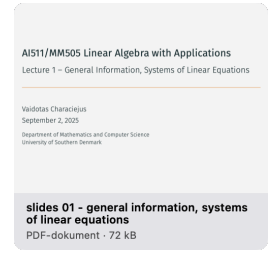

Nathaniel Johnston
Introduction to Linear and Matrix Algebra
Springer
introduction-to-linear-and-matrix-algebra-by-nathaniel-johnston
PDF-dokument · 16,4 MB

topics for this course


- Systems of linear equations.
- Matrix operations, inverses, determinants.
- Vector spaces, basis, coordinates, linear independence.
- Linear transformations, eigenvalue problems, diagonalisation.
- Inner product and orthogonality.

outdated


AI511/MM505 Linear Algebra with Applications
Lecture 1 – General information; systems of linear equations
slides 01 - general information, systems of linear equations
PDF-dokument · 74 kB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Lecture 1 – General information; Systems of Linear Equations
slides 01 - general information, systems of linear equations
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lecture 01

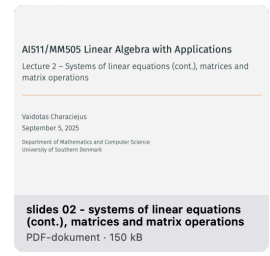
5 ECTS theory
- take home exam

2.5 ECTS Application
- explained by Henry

Systems of linear equations

use calculus to find linear equation (optimization)

outdated


AI511/MM505 Linear Algebra with Applications
Lecture 2 – Systems of linear equations (cont.), matrices and matrix operations
slides 02 - systems of linear equations (cont.), matrices and matrix operations
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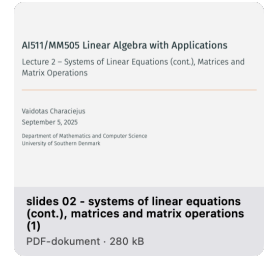
AI511/MM505 Linear Algebra with Applications
Lecture 2 – Systems of Linear Equations (cont.), Matrices and Matrix Operations
slides 02 - systems of linear equations (cont.), matrices and matrix operations (1)
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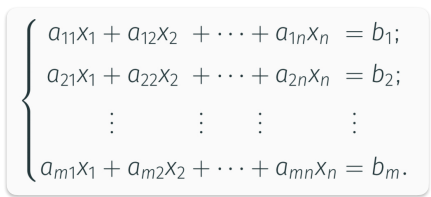
Lecture 02

general system of linear equation...


$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1;\\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2;\\ \quad\vdots \qquad\quad \vdots \qquad\qquad \vdots \qquad \vdots\\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

augmented matrix ↰

a_{ij} & b_i are entries

this can be rewritten to

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1\\ a_{21} & a_{22} & \cdots & a_{2n} & b_2\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ a_{n1} & a_{n2} & & a_{mn} & b_n \end{array}\right]$$

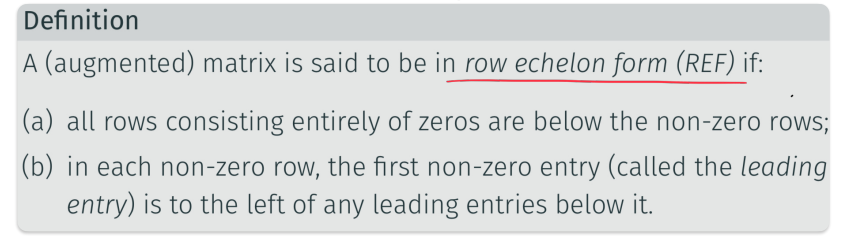
Denotion

$\mathbb{R}^{m\times n}$

$(a_{ij})_{1\le i\le m,\ 1\le j\le n}$

now use elementary row operations
(Gauss reduction)

Definition of an augmented matrix

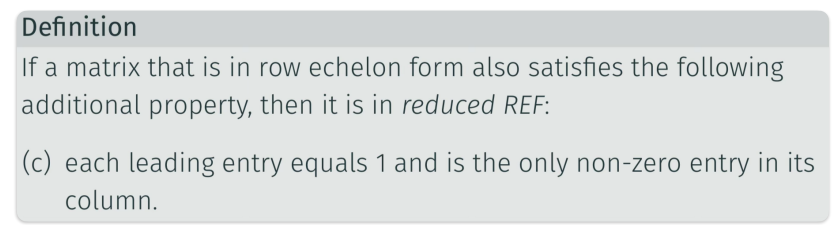

Definition

A (augmented) matrix is said to be in *row echelon form (REF)* if:

(a) all rows consisting entirely of zeros are below the non-zero rows;
(b) in each non-zero row, the first non-zero entry (called the *leading entry*) is to the left of any leading entries below it.

if REF
use back substitution
elif reduced REF
read the answer

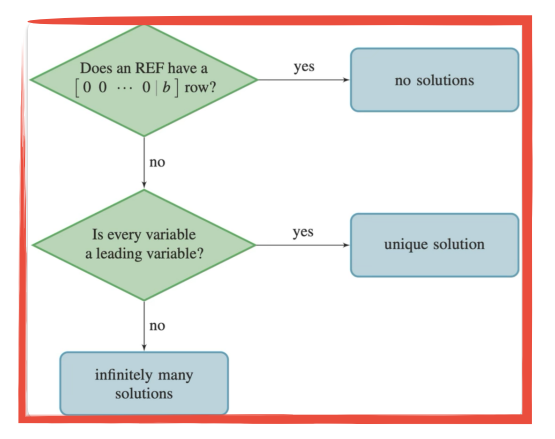
using Gauss-Jordan ↓ Definition of reduced REF


Definition

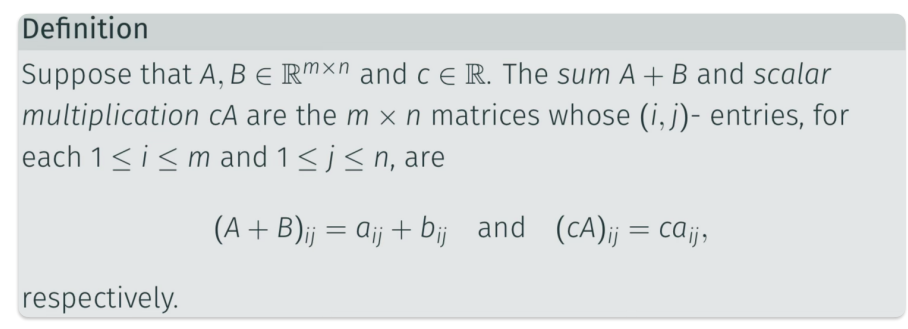
If a matrix that is in row echelon form also satisfies the following additional property, then it is in *reduced REF*:

(c) each leading entry equals 1 and is the only non-zero entry in its column.

if the number of leading entries is less than the number of variables then there are infinite amount of solutions


Does an REF have a [0 0 ⋯ 0 | b] row?
yes
no solutions
no
Is every variable a leading variable?
yes
unique solution
no
infinitely many solutions

Matrix addition & scalar multiplication

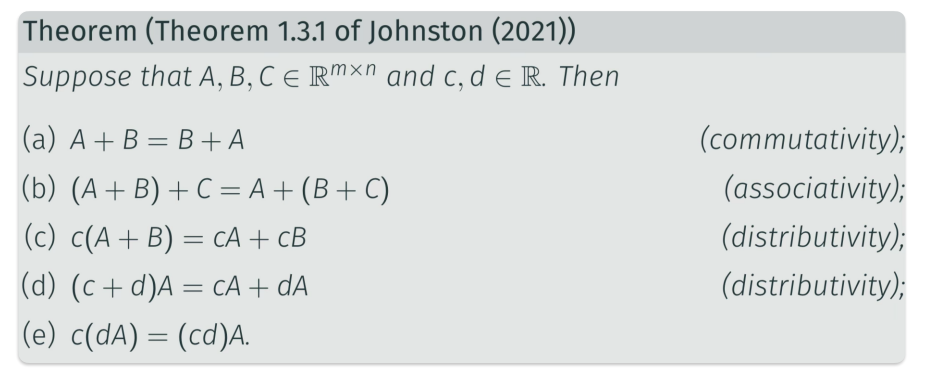

Definition

Suppose that $A, B \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}$. The *sum* $A + B$ and *scalar multiplication* cA are the $m \times n$ matrices whose (i, j)- entries, for each $1 \le i \le m$ and $1 \le j \le n$, are

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad\text{and}\quad (cA)_{ij} = ca_{ij},$$

respectively.

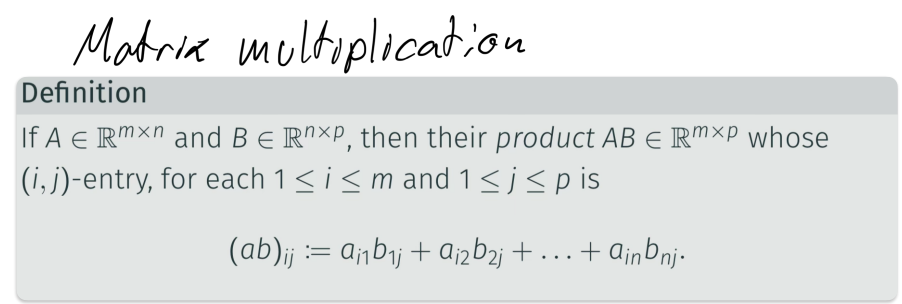
↓ common rules for matrix addition & scalar multiplication


Theorem (Theorem 1.3.1 of Johnston (2021))

Suppose that $A, B, C \in \mathbb{R}^{m\times n}$ and $c, d \in \mathbb{R}$. Then

(a) $A + B = B + A$ *(commutativity);*
(b) $(A + B) + C = A + (B + C)$ *(associativity);*
(c) $c(A + B) = cA + cB$ *(distributivity);*
(d) $(c + d)A = cA + dA$ *(distributivity);*
(e) $c(dA) = (cd)A$.

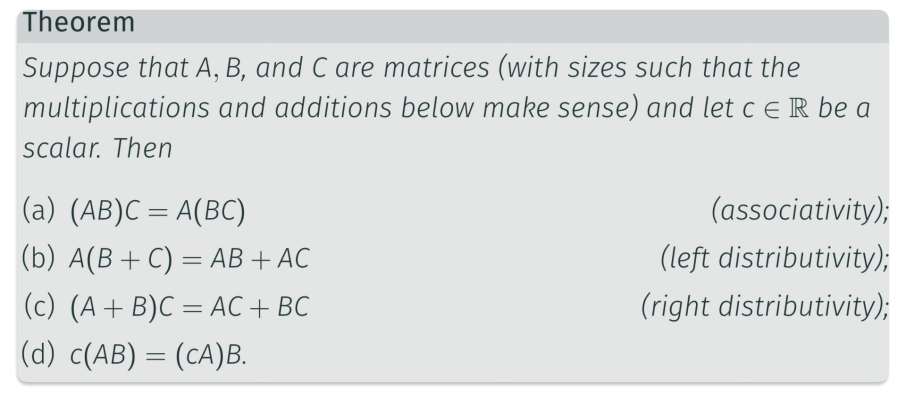
Matrix multiplication


Definition

If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, then their *product* $AB \in \mathbb{R}^{m\times p}$ whose (i, j)-entry, for each $1 \le i \le m$ and $1 \le j \le p$ is

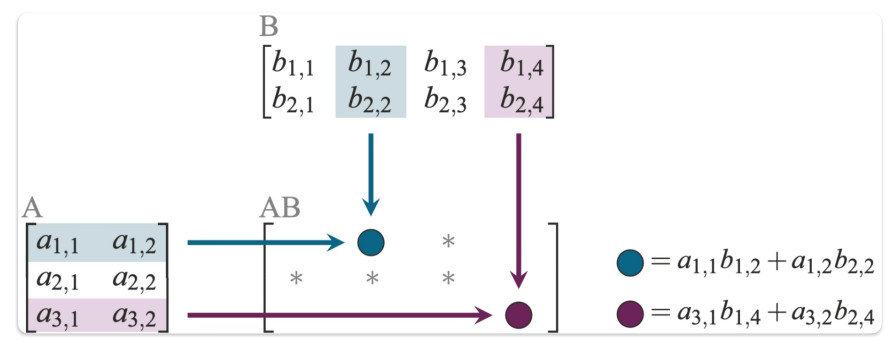
$$(ab)_{ij} := a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{in}b_{nj}.$$

↓ common rules for matrix multiplication ↓

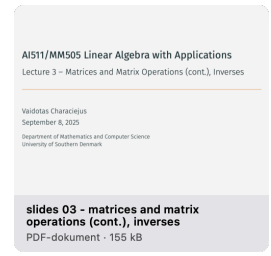

Theorem

Suppose that A, B, and C are matrices (with sizes such that the multiplications and additions below make sense) and let $c \in \mathbb{R}$ be a scalar. Then

(a) $(AB)C = A(BC)$ *(associativity);*
(b) $A(B + C) = AB + AC$ *(left distributivity);*
(c) $(A + B)C = AC + BC$ *(right distributivity);*
(d) $c(AB) = (cA)B$.


B
$b_{1,1}$ $b_{1,2}$ $b_{1,3}$ $b_{1,4}$
$b_{2,1}$ $b_{2,2}$ $b_{2,3}$ $b_{2,4}$
A
$a_{1,1}$ $a_{1,2}$
$a_{2,1}$ $a_{2,2}$
$a_{3,1}$ $a_{3,2}$
AB
$= a_{1,1}b_{1,2} + a_{1,2}b_{2,2}$
$= a_{3,1}b_{1,4} + a_{3,2}b_{2,4}$

outdated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Lecture 3 – Matrices and Matrix Operations (cont.), Inverses
slides 03 - matrices and matrix operations (cont.), inverses
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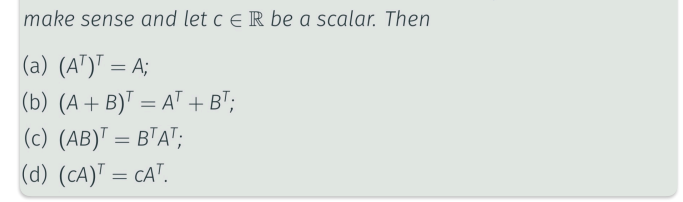
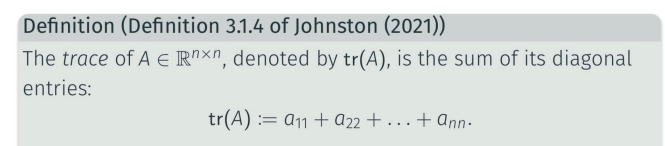
transposing matrices


Theorem (Theorem 1.3.4 of Johnston (2021))

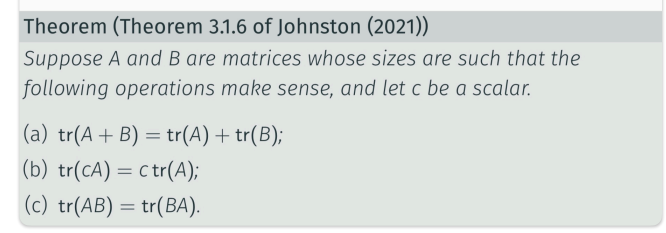
Let A and B be matrices with sizes such that the operations below make sense and let $c \in \mathbb{R}$ be a scalar. Then

(a) $(A^T)^T = A$;
(b) $(A + B)^T = A^T + B^T$;
(c) $(AB)^T = B^TA^T$;
(d) $(cA)^T = cA^T$.


Definition (Definition 3.1.4 of Johnston (2021))

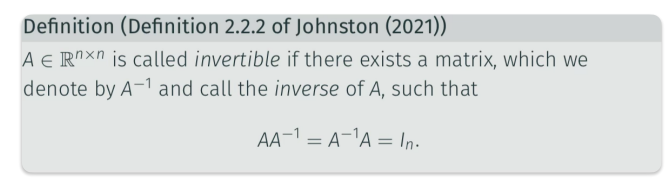
The *trace* of $A \in \mathbb{R}^{n\times n}$, denoted by $\mathrm{tr}(A)$, is the sum of its diagonal entries:

$$\mathrm{tr}(A) := a_{11} + a_{22} + \ldots + a_{nn}.$$


Theorem (Theorem 3.1.6 of Johnston (2021))

Suppose A and B are matrices whose sizes are such that the following operations make sense, and let c be a scalar.

(a) $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$;
(b) $\mathrm{tr}(cA) = c\,\mathrm{tr}(A)$;
(c) $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.


Definition (Definition 2.2.2 of Johnston (2021))

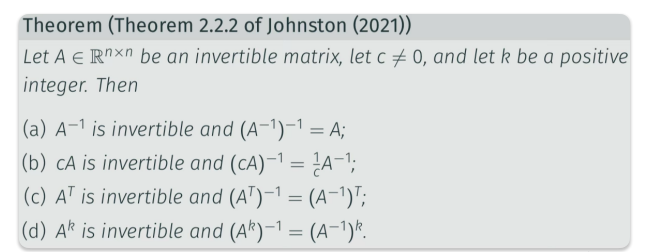
$A \in \mathbb{R}^{n\times n}$ is called *invertible* if there exists a matrix, which we denote by A^{-1} and call the *inverse* of A, such that

$$AA^{-1} = A^{-1}A = I_n.$$

use inverse matrix to solve

$A = \begin{bmatrix} 2 & -5\\ -1 & 3\end{bmatrix}$ $A^{-1} = \begin{bmatrix} 3 & 5\\ 1 & 2\end{bmatrix}$

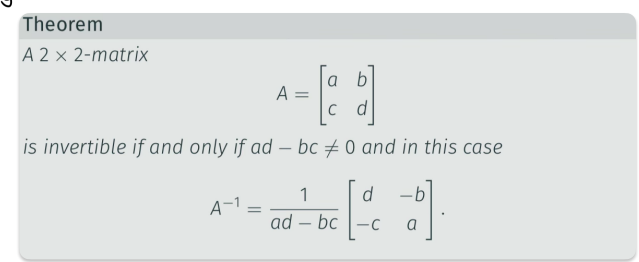
$A^{-1}A = \begin{bmatrix} 1 & 0\\ 0 & 1\end{bmatrix}$ ← unige for A!


Theorem (Theorem 2.2.2 of Johnston (2021))

Let $A \in \mathbb{R}^{n\times n}$ be an invertible matrix, let $c \neq 0$, and let k be a positive integer. Then

(a) *A^{-1} is invertible and $(A^{-1})^{-1} = A$;*
(b) *cA is invertible and $(cA)^{-1} = \frac{1}{c}A^{-1}$;*
(c) *A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$;*
(d) *A^k is invertible and $(A^k)^{-1} = (A^{-1})^k$.*

{ common rules for invertable matrices

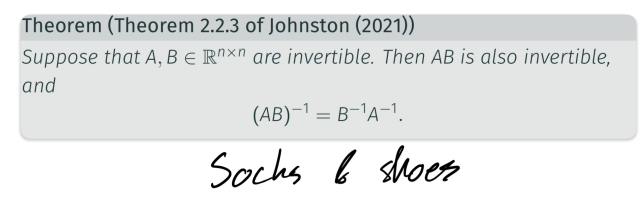

Theorem

A 2 × 2-matrix

$$A = \begin{bmatrix} a & b\\ c & d\end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$ and in this case

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b\\ -c & a\end{bmatrix}.$$

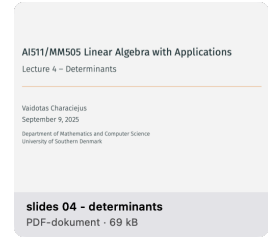

Theorem (Theorem 2.2.3 of Johnston (2021))

Suppose that $A, B \in \mathbb{R}^{n\times n}$ are invertible. Then AB is also invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Socks & shoes

inversion of a matrix is only defined for a $\mathbb{R}^{n\times n}$ (square) matrices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Lecture 4 – Determinants
slides 04 - determinants
PDF-dokument · 69 kB

if A is invertable then A's rREF $= I_n$

find the inverse:

$A \xrightarrow{ERO} B$
use elementary row operation on both A & I_n
$I_n \xrightarrow{ERO} E$

note: left multiplication ↓ $EA = B$

example:

$A = \begin{bmatrix} 1 & 2 & 3\\ 2 & 5 & 3\\ 1 & 0 & 8\end{bmatrix}$

find inverse of A by bringing the augmented matrix

Gauss-Jordan

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0\\ 2&5&3&0&1&0\\ 1&0&8&0&0&1 \end{array}\right]$$

$$\xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0\\ 0&1&-3&-2&1&0\\ 1&0&8&0&0&1 \end{array}\right]$$

$$\xrightarrow{R_3 - R_1} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0\\ 0&1&-3&-2&1&0\\ 0&-2&5&-1&0&1 \end{array}\right]$$

$$\xrightarrow{R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0\\ 0&1&-3&-2&1&0\\ 0&0&-1&-5&2&1 \end{array}\right]$$

$$\xrightarrow{1 - R_3} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0\\ 0&1&-3&-2&1&0\\ 0&0&1&5&-2&-1 \end{array}\right]$$

$$\xrightarrow{R_2 + 3R_3} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0\\ 0&1&0&13&-5&-3\\ 0&0&1&5&-2&-1 \end{array}\right]$$

$$\xrightarrow{R_1 - 3R_3} \left[\begin{array}{ccc|ccc} 1&2&0&-14&6&3\\ 0&1&0&13&-5&-3\\ 0&0&1&5&-2&-1 \end{array}\right]$$

$$\xrightarrow{R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1&0&0&-40&16&9\\ 0&1&0&13&-5&-3\\ 0&0&1&5&-2&-1 \end{array}\right]$$

A is invertable $\Longleftrightarrow$ $Ax = 0$ has a unige solution

Step 1
$Ax = 0$
$BAx = B0$
$I_nx = 0$
$x = 0$

Step 2
$A^{-1}A = AA^{-1} = I_n$
$BA = I_n$
$B\underbrace{AA^{-1}}_{I_n} = I_nA^{-1}$
$B = A^{-1}$

for every $b \in \mathbb{R}^{n\times 1}$
where $A \in \mathbb{R}^{n\times n}$ & A is invertable

$Ax = b$
$\Downarrow$
$x = A^{-1}b$

$Ax = b$
$A(A^{-1}b) = b$
$\underbrace{AA^{-1}}_{I_n}b = b$
$b = b$

$A\tilde{x} = b$
A^{-1}

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Lecture 5 – Determinants; Euclidean Vector Spaces
slides 05 - determinants, euclidean vector spaces
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Some special matrices

$Ax = b$ $\quad A = PDP^{-1}$
$x = A^{-1}b$ $\quad A^2 = PD^2P^{-1}$
$AB = I_n$ $\quad A^{-1} = PD^{-1}P^{-1}$
$BA = I_n$

diagonal matrices

$\begin{bmatrix} 1&0&0\\ 0&2&0\\ 0&0&3\end{bmatrix} = \mathrm{diag}(1,2,3)$

$\mathrm{diag}(a_1 \dots a_n)\,\mathrm{diag}(b_1 \dots b_n) = \mathrm{diag}(a_1b_1 \dots a_nb_n)$

$D^{-1} = (d_1^{-1} \dots d_n^{-1})$
(only if $d_i \neq 0$)

$D^2 = (d_1^2 \dots d_n^2)$

triangular matrices

upper $= \begin{bmatrix} 1&2&3\\ 0&4&5\\ 0&0&6\end{bmatrix}$

lower $= \begin{bmatrix} 1&0&0\\ 2&3&0\\ 4&5&6\end{bmatrix}$

a) the transpose of a lower tri. matrix is upper
b) the product of 2 upper tri-matrices are upper

Determinants

recall that for
$A = \begin{bmatrix} a & b\\ c & d\end{bmatrix}$
$\det(A) = ad - bc$
if $\det(A) \neq 0$ then A is invertable

minors & cofactors

$A = \begin{bmatrix} a_{11} & \cancel{a_{12}}\\ \cancel{a_{21}} & a_{22}\end{bmatrix}$ $\quad M_{11} = \det([a_{22}])$

$C_{ij} = (-1)^{i+j}M_{ij}$

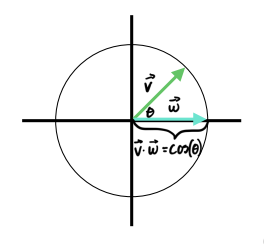
note that $\det(A + B) \neq \det(A) + \det(B)$ in all instances

if matrix A is triangular
$\det(A) = \prod_{i=1}^{n} a_{ii}$

for matrix $A \in \mathbb{R}^{n\times n}$
(a) by multiplying a row with $k \in \mathbb{R}$, then $\det(B) = k \cdot \det(A)$
(b) interchanging two rows, then $\det(B) = -\det(A)$
(c) adding a multiple one row to another, then $\det(B) = \det(A)$

A matrix A is invertible if and only if $\det(A) \neq 0$

Vectors



observe that
$\vec{v}\cdot\vec{w} = \vec{v}^T\vec{w} = \vec{v}\,\vec{w}^T$
$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 \dots v_n^2}$

euclidean norm

for $\vec{v} \in \mathbb{R}^n$ & $c \in \mathbb{R}$
(a) $\|c\vec{v}\| = |c|\|\vec{v}\|$

for any $v \neq 0$ vectors
$\vec{v} = \|\vec{v}\|\vec{u}$
where $\vec{u}$ is a unit vector

the standard basis of in $\mathbb{R}^n$

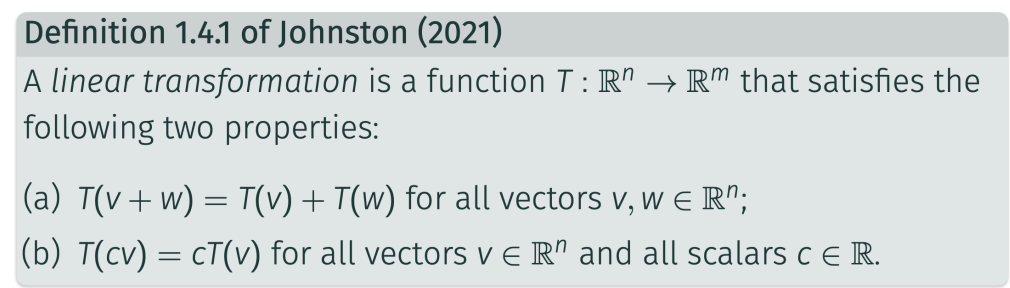
$e_1 = \begin{bmatrix} 1\\ 0\\ \vdots\\ 0\end{bmatrix}, e_2 = \begin{bmatrix} 0\\ 1\\ \vdots\\ 0\end{bmatrix}, \dots, e_n \begin{bmatrix} 0\\ 0\\ \vdots\\ 1\end{bmatrix}.$

and so..
any vector in $\mathbb{R}^n$ can be written in terms of the vectors $e_1, e_2, \dots, e_n$ in the following way

$v = (v_1e_1 + v_2e_2 + \dots + v_ne_n)$

Propeties of matrix transformations

linear transformations
satisfies the following properties
(a) $T(v + w) = T(v) + T(w)$ for all vectors $v, w \in \mathbb{R}^n$


Definition 1.4.1 of Johnston (2021)

A *linear transformation* is a function $T : \mathbb{R}^n \to \mathbb{R}^m$ that satisfies the following two properties:

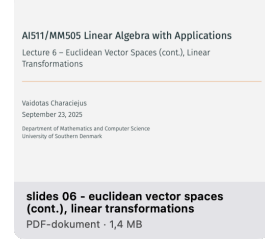
(a) $T(v + w) = T(v) + T(w)$ for all vectors $v, w \in \mathbb{R}^n$;
(b) $T(cv) = cT(v)$ for all vectors $v \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$.

transformation is linear if
scaling then mapping =
mapping then scaling

Standard matrix of a linear transformation
a function $T: \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there exists a matrix $[T] \in \mathbb{R}^{m\times n}$ such that
$T(v) = [T]v$ for all $v \in \mathbb{R}^n$

note:
$[T] = [T(e_1)\ T(e_2) \dots T(e_n)]$

example
$T(v_1, v_2) = (v_1 + 2v_2, 3v_1 + 4v_2)$
$T(e_1) = T(1,0) = (1,3)$
$T(e_2) = T(0,1) = (2,4)$
$[T] = \begin{bmatrix} 1 & 2\\ 3 & 4\end{bmatrix}$
$[T]\cdot\begin{bmatrix} v_1\\ v_2\end{bmatrix} = \begin{bmatrix} 1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix} v_1\\ v_2\end{bmatrix} = \begin{bmatrix} v_1 & 2v_2\\ 3v_1 & 4v_2\end{bmatrix}$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Lecture 6 – Euclidean Vector Spaces (cont.), Linear Transformations
slides 06 - euclidean vector spaces (cont.), linear transformations
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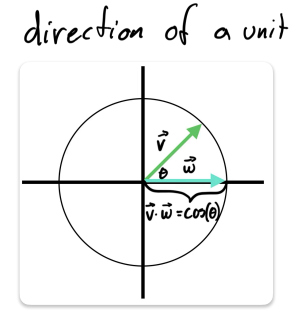
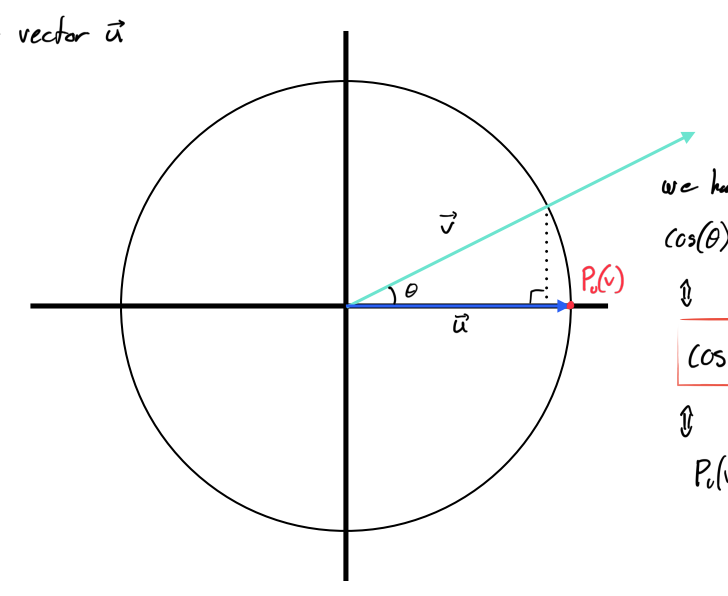
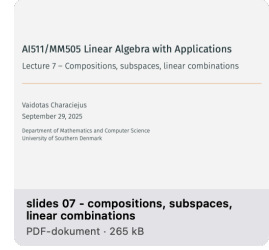
Orthogonal projection onto the direction of a unit vector $\vec{u}$





we know that
$\cos(\theta) = \frac{ad}{hyp} = \frac{\|P_u(v)\|}{\|v\|}$
$\Updownarrow$
$\boxed{\cos(\theta)\cdot\|v\| = \|P_u(v)\|}$
$\Updownarrow$
$P_u(v) = \|v\|\cdot\cos(\theta)\cdot u$
$= \|v\|\cdot\frac{v\cdot u}{\|v\|\,\|u\|}\cdot u$
$= (v\cdot u)\,u$
$= u^Tv\,u$
$= uu^Tv$

Dot product
$\frac{w\cdot v}{\|w\|\cdot\|v\|} = \cos(\theta)$
so for unit vectors
$w\cdot v = \cos(\theta)$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Lecture 7 – Compositions, subspaces, linear combinations
slides 07 - compositions, subspaces, linear combinations
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Lecture 8 – Subspaces, Linear Combinations, Bases and Coordinates, Dimension
slides 08 - subspaces, linear combinations, bases and coordinates, dimension
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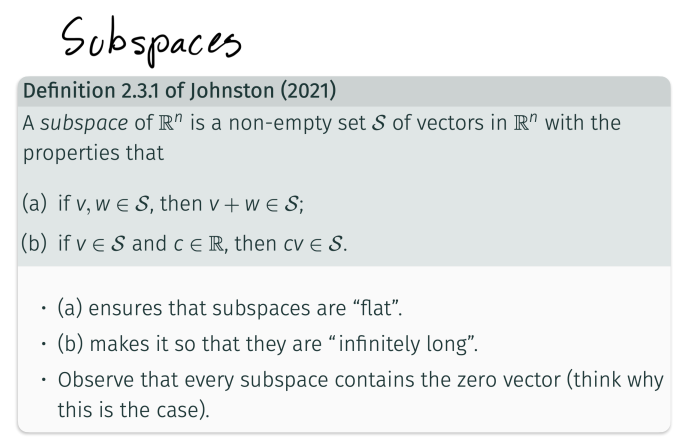
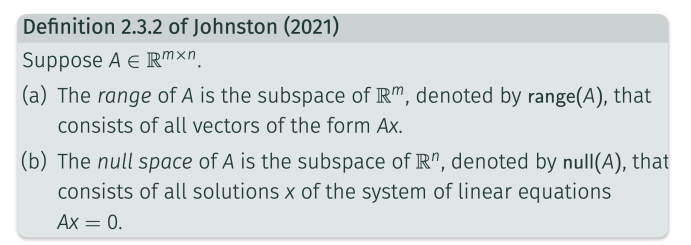
Subspaces


Definition 2.3.1 of Johnston (2021)

A *subspace* of $\mathbb{R}^n$ is a non-empty set $\mathcal{S}$ of vectors in $\mathbb{R}^n$ with the properties that

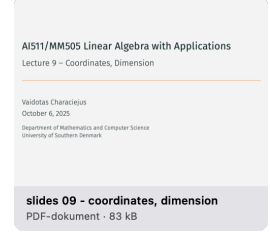
(a) if $v, w \in \mathcal{S}$, then $v + w \in \mathcal{S}$;
(b) if $v \in \mathcal{S}$ and $c \in \mathbb{R}$, then $cv \in \mathcal{S}$.

- (a) ensures that subspaces are "flat".
- (b) makes it so that they are "infinitely long".
- Observe that every subspace contains the zero vector (think why this is the case).


Definition 2.3.2 of Johnston (2021)

Suppose $A \in \mathbb{R}^{m\times n}$.

(a) The *range* of A is the subspace of $\mathbb{R}^m$, denoted by range(A), that consists of all vectors of the form Ax.
(b) The *null space* of A is the subspace of $\mathbb{R}^n$, denoted by null(A), that consists of all solutions x of the system of linear equations $Ax = 0$.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Lecture 10 – Change of Basis, Fundamental Subspaces, Rank and Nullity
slides 10 - change of basis, fundamental subspaces, rank and nullity
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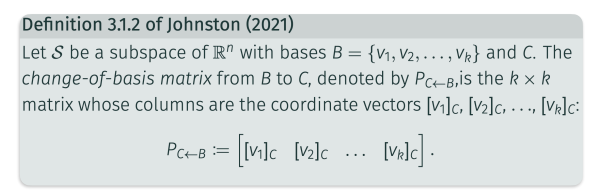
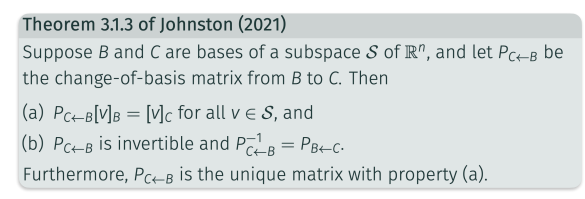
Chang-of-basis matrix


Definition 3.1.2 of Johnston (2021)

Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ with bases $B = \{v_1, v_2, \dots, v_k\}$ and C. The *change-of-basis matrix* from B to C, denoted by $P_{C\leftarrow B}$, is the $k \times k$ matrix whose columns are the coordinate vectors $[v_1]_C, [v_2]_C, \dots, [v_k]_C$:

$$P_{C\leftarrow B} := \big[[v_1]_C \quad [v_2]_C \quad \dots \quad [v_k]_C\big].$$


Theorem 3.1.3 of Johnston (2021)

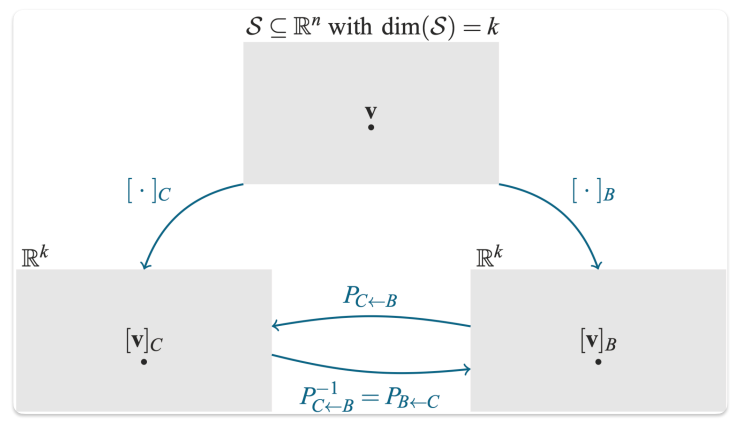
Suppose B and C are bases of a subspace $\mathcal{S}$ of $\mathbb{R}^n$, and let $P_{C\leftarrow B}$ be the change-of-basis matrix from B to C. Then

(a) $P_{C\leftarrow B}[v]_B = [v]_C$ for all $v \in \mathcal{S}$, and
(b) $P_{C\leftarrow B}$ is invertible and $P^{-1}_{C\leftarrow B} = P_{B\leftarrow C}$.

Furthermore, $P_{C\leftarrow B}$ is the unique matrix with property (a).

for $P_{C\leftarrow B} \in \mathbb{R}^{k\times k}$

$P_{C\leftarrow B}[v]_B = [v]_C$

also..
$P_{C\leftarrow B}^{-1} = P_{B\leftarrow C}$